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Comparative Evaluation of Transformer-Based Models for Automated Short-Answer Grading

Temidayo Oluwatosin Omotehinwa
Department of Mathematics and Computer Science
Federal University of Health Sciences
Otukpo, Nigeria
temidayo.omotehinwa@fuhso.edu.ng

Rock Adeiza Ezekiel
Directorate of Information Technology and Resource Centre (ITRC)
Kogi State Polytechnic
Lokoja, Kogi State
rock.ezekiel@kogistatepolytechnic.edu.ng

Monday Onoja
Department of Applied Informatics
Comenius University Bratislava
Bratislava, Slovakia
monday.onoja@fmph.uniba.sk

Emmanuel Chizoba Omeye
Department of Mathematics and Computer Science
Federal University of Health Sciences
Otukpo, Nigeria
eminentemma.chizoba@gmail.com

Yakubu Aliyu Ibrahim
Department of Mathematics and Computer Science
Federal University of Health Sciences
Otukpo, Nigeria
talktoibro80@gmail.com

Abstract*—The grading of open-ended questions in educational settings is a labour-intensive and subjective process, often leading to inconsistencies and biases in scoring. Conventional automated and manual grading systems have faced challenges in accurately interpreting the complexity and variability of student responses. This study addressed these challenges by developing a transformer-based scoring approach that leverages advanced artificial intelligence and semantic similarity techniques to provide consistent, accurate, and efficient grading of open-ended questions. The study evaluated the performance of MiniLM, BERT, RoBERTa, and SBERT in grading short-answer questions based on correlation, agreement, semantic sensitivity, and computational efficiency. Results indicate that SBERT exhibits the highest semantic sensitivity across different questions (Q1: 0.726, Q2: 0.881, Q3: 0.798) and strong correlation with MiniLM (Pearson: 0.909, Spearman: 0.908). MiniLM provides a competitive balance between efficiency and accuracy, processing responses in 80.06 seconds compared to RoBERTa's 1359.64 seconds, making MiniLM 16.98 times faster and substantially more efficient for grading large volumes of responses in real-time or large-scale grading scenarios. Score distribution analysis revealed that SBERT and MiniLM produce similar grading patterns, while RoBERTa demonstrates the highest variance. BERT and RoBERTa demonstrate near-random agreement (-0.020), as reflected in Cohen's Kappa scores which indicates inconsistencies. These findings emphasize the importance of model selection in automated grading, with SBERT emerging as the most robust in capturing semantic similarity. The proposed approach demonstrates the potential to alleviate educators' grading burden with transformer-based scoring while enhancing the fairness and reliability of student evaluations.

Keywords*— Automated Grading, Sentence-Transformers, Semantic Similarity, Short-Answer Assessment, Machine Learning in Education, Natural Language Processing

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I. INTRODUCTION

The assessment of student performance is a very important component of the educational process, as it provides valuable insights into learning outcomes, knowledge retention, and conceptual understanding. Traditional assessment methods, particularly those involving open-ended questions, present significant challenges for human evaluators in terms of scalability, subjectivity, and grading consistency [1], [2]. These challenges can result in inconsistencies and a lack of fairness in evaluations [3], [4]. Unlike multiple-choice or fill-in-the-blank assessments, which can be graded efficiently using rule-based algorithms, evaluating open-ended responses requires deeper semantic understanding, making manual grading both time-consuming, labour-intensive and prone to inconsistencies [5], [6], [7].

Recent advancements in artificial intelligence (AI) and natural language processing (NLP) have revolutionized various domains [8], [9], [10], including educational assessment [11], [12], [13]. AI-driven automated grading systems have emerged as a promising solution to alleviate the burdens of manual grading while enhancing accuracy, efficiency, and fairness [14], [15]. Among AI-based methods, transformer-based models, such as BERT (Bidirectional Encoder Representations from Transformers) and Sentence-BERT (SBERT), have demonstrated state-of-the-art performance in semantic text similarity and comprehension tasks [16], [17], [18], [19]. These models leverage deep learning techniques to capture intricate language structures, enabling them to assess responses based on semantic similarity rather than mere keyword matching. Despite their potential, selecting the most suitable transformer model for automated grading remains a critical challenge, as different models exhibit varying levels of accuracy, agreement, and computational efficiency.

The application of transformer-based models in grading open-ended questions holds significant promise. These models can potentially address the limitations of traditional and earlier automated grading systems by offering a more sophisticated analysis of student responses by capturing contextual meaning, paraphrased expressions, and subtle linguistic variations. They can analyse students' written responses, evaluate the content and structure, and generate scores that align with human evaluators' standards. Despite these potential benefits, the integration of AI in grading open-ended questions presents several challenges such as understanding context, handling ambiguity, lack of domain specific knowledge and deployment [20], [21]. Ensuring the reliability, fairness, and transparency of AI-driven grading systems is paramount.

This study aims to evaluate the performance of several transformer-based models: MiniLM, BERT, RoBERTa, and SBERT in the grading of short-answer questions. The models are assessed based on their correlation in model scores, agreement metrics, semantic sensitivity, and computational efficiency. Specifically, the study seeks to determine which model offers the best balance between accuracy and processing speed while maintaining reliable grading patterns.

By addressing the challenges of inconsistency and subjectivity in manual grading, this research highlights the potential of transformer-based scoring approaches to reduce grading time, improve fairness and reliability of student evaluations. Furthermore, this study provides empirical evidence on the reliability and effectiveness of transformer-based models for automated evaluation, paving the way for broader adoption in computer-assisted learning, remote education, and standardized testing.

II. Related Works

Numerous methodologies have been proposed for automated short-answer grading systems. This section reviews relevant studies that focus on the automatic evaluation of short-answer responses.

The study by [22] developed a framework for automatic grading of short-answers. The framework combined semantic features based on BERT and lexical features using TF-IDF (Term Frequency-Inverse Document Frequency) for improved accuracy in grading. A unique part of the framework is the support for non-English dataset. The experimental results showed that the framework named GradeAid achieved a competitive performance with a root mean squared error (RMSE) of 0.25.

The study by [23] focused on the automatic grading of responses to questions using semantic similarity in a low resource language; Arabic. Local and hybrid local-global weighting technique was applied to a dataset containing 2,133 model-student answer pairs. The evaluation results revealed that the hybrid local-global weighting achieved a superior performance with a F1-score of 82.82% and RMSE of 0.798.

In a bid to autonomously and accurately evaluate short-answer questions [24], hybridized multiple tokenizers (DistilBERT, MPNet, XLNet and RoBERTa) with BERT. The evaluation of these models revealed an accuracy up to 99.6%. However, the generalizability across diverse subject is yet to be established.

A study by [25] introduced GRAD-AI, an AI-powered grading tool for computer programming assignments that combines automation with teacher involvement. Using techniques like the Halstead Complexity Measure, TF-IDF, Abstract Syntax Tree processing, and K-means clustering, it ensures accurate, unbiased, and timely feedback. Experimental results showed its effectiveness in delivering precise assessments and real-time feedback, fostering interactive learning.

Reference [26] focused on the assessment of Ai model's performance on open-ended medical questions. In the study, 21 open-ended certification examination questions on head and neck were used to test ChatGPT's performance. The responses from the language model were evaluated based on concordance, validity, safety and competency (CVSC). After three trials ChatGPT achieved mean passing score of 75% and performed better when guided with prompts. High concordance (92.06%) and satisfactory validity were also recorded. However, the study noted that ChatGPT sometimes provided partially correct responses and generated misleading information. In a similar study [27] an investigation into the effectiveness of fine-tuned GPT-3.5 for the grading of students responses to science education questions was carried out in a bid to address the limitation of direct pre-trained model usage due to differences in student language pattern and the need for contextual understanding. The performance of the fine-tuned GPT-3.5 was compared to fine-tuned BERT. The result revealed that GPT3.5 achieved a superior performance over BERT with a mean scoring accuracy of 9.1% across tasks.

The feasibility of GPT and Llama for automatic grading of short-answer was investigated in this study [28]. The experimental setup included zero-shot, few-shot, fine-tuned variants of GPT and Llama, and supervised (BERT and SVM)

models. The results revealed that the LLMs performance for zero-shot and few-shot settings was very poor, rendering them impractical for accurate assessment without additional training. The LLMs performance matched the supervised models after fine-tuning. The efficiency trade-offs suggest that LLMs are not yet feasible for short-answer grading. Also, a recent study [29] which investigated the effectiveness of Chatgpt-4 compared to human in scoring answers and providing feedback on dental students histology assignment found that Chatgpt-4 was superior in identifying mistakes and providing clear improvement suggestion. However, the students expressed greater comfort in human tutor scoring and feedback.

The study by [30] hybridized word level text preprocessing algorithm, artificial neural network (ANN), text similarity algorithm and heuristic decision rules for scoring students' answer in Polish. The proposed approach achieved a high grading accuracy. However, the false negatives indicate the need for further refinements.

It can be observed from the related work that existing studies on automated short-answer grading have made significant strides but still present notable gaps. Many frameworks, such as GradeAid [22] and GRAD-AI [25], integrate AI-driven approaches but rely heavily on traditional lexical features such as TF-IDF or structural analysis (Abstract Syntax Trees), limiting their ability to capture deep semantic meaning. While studies leveraging transformer models, such as those by [24] and [28], highlight their high accuracy, concerns remain about their generalizability across diverse subjects, languages, and question types. Additionally, research into large language models (LLMs) like GPT-3.5 and Llama [27], [28] demonstrates that their zero-shot and few-shot performance is inadequate for grading, requiring extensive fine-tuning for effective assessment. Furthermore, comparative analyses of different transformer-based models for short-answer grading are scarce, with most studies focusing on a single model rather than evaluating multiple architectures in terms of accuracy, efficiency, and agreement.

Given these limitations, this study justifies its contributions by conducting a comparative analysis of MiniLM, BERT, RoBERTa, and SBERT to determine the most effective transformer-based model for short-answer grading. Unlike previous studies that focus on either lexical features or a single transformer model, this research evaluates multiple models across critical metrics, including semantic sensitivity, correlation, agreement, and computational efficiency. By doing so, it provides deeper insights into model selection for automated grading, identifying an optimal balance between accuracy and efficiency.

III. Methodology

A. Data Collection

The dataset used in this study comprises responses from 225 students offering COS101: Introduction to Computing Science at the Federal University of Health Sciences, Otukpo, Benue State, Nigeria [31]. The students answered three open-ended questions: (Q1) What was the significance of Ada Lovelace's work on the Analytical Engine? (Q2) What contribution did John von Neumann make to computer architecture? (Q3) What was the contribution of John Presper Eckert and John W. Mauchly to computer development? The data was collected via Google Forms and stored in a CSV file containing four columns: Matric Number, Q1, Q2, and Q3. The columns Q1, Q2, and Q3 holds the students' responses to each of the questions asked. Each response was evaluated against predefined correct answers to determine semantic similarity and assign scores.

B. Sentence Embedding Models

Four transformer-based sentence embedding models were employed to measure semantic similarity between student responses and the correct answers:

- MiniLM (paraphrase-MiniLM-L6-v2): is a lightweight yet powerful refined version of large transformer models like BERT and RoBERTa, designed for efficiency without compromising performance. By leveraging deep self-attention distillation, it effectively compresses the knowledge of larger models into a more compact form while preserving their core capabilities [32]. By optimizing the number of attention heads and hidden layers, MiniLM achieves fast inference speeds and efficient memory usage, making it ideal for real-time NLP applications.

- BERT (Bidirectional Encoder Representations from Transformers) (bert-base-nli-mean-tokens): is a transformer-based model developed by Google AI that revolutionized natural language processing. Unlike traditional models that process text sequentially, BERT leverages deep bidirectional representations, allowing it to jointly consider both left and right context across all layers [33]. This bidirectional approach enables a more comprehensive understanding of language nuances, making BERT highly effective for tasks such as question answering, sentiment analysis, and text classification.

- RoBERTa (Robustly Optimized BERT Pretraining Approach) (sentence-transformers/all-roberta-large-v1): is an advanced refinement of BERT, developed by Facebook AI [34], that enhances performance through key optimizations. By training on larger mini-batches, eliminating the next-sentence prediction objective, and employing dynamic masking, RoBERTa achieves a deeper understanding of contextual relationships in text [34]. These improvements enable it to outperform BERT across various natural language processing tasks by capturing richer linguistic patterns while maintaining greater efficiency [35], [36], [37].

- Sentence-BERT (SBERT) (all-MiniLM-L12-v2): is an enhanced variant of BERT designed to generate semantically rich sentence embeddings using Siamese and triplet network architectures [38]. This approach not only captures contextual meaning more effectively but also drastically improves efficiency in tasks such as semantic similarity search and clustering. Unlike the

original BERT model, which requires computationally expensive cross-encoding for pairwise comparisons, SBERT enables quick and scalable similarity computations by precomputing fixed-length vector representations [38]. As a result, it delivers high accuracy while significantly reducing processing time, making it particularly well-suited for large-scale natural language understanding applications.

C. Semantic Similarity Computation

Each model generates vector embeddings for both student responses and predefined correct answers. The similarity between these embeddings is computed using cosine similarity as in (1). A and B are the vector embeddings of the student response and correct answer, respectively.

$$Cosine\ Similarity(A, B) = \frac{A.B}{\|A\|\|B\|} \quad (1)$$

Where:

$A.B$ is the dot product of the two vectors and it is computed as in (2).

$$A.B = \sum_{i=1}^{n} A_i B_i \quad (2)$$

$\|A\|$ and $\|B\|$ are the Euclidean norms (magnitudes) of A and B computed as defined in (3) and (4).

$$\|A\| = \sqrt{\sum_{i=1}^{n} A_i^2} \quad (3)$$

$$\|B\| = \sqrt{\sum_{i=1}^{n} B_i^2} \quad (4)$$

Cosine similarity ranges from -1 (completely opposite) to 1 (identical), with 0 indicating no similarity.

D. Grading Framework

A scoring rubric was established based on similarity thresholds:

- Similarity ≥ 0.8 → Score: 5 (Very similar)
- 0.6 ≤ Similarity < 0.8 → Score: 3 (Moderately similar)
- 0.4 ≤ Similarity < 0.6 → Score: 1 (Somewhat similar)
- Similarity < 0.4 → Score: 0 (Not similar)

Each model assigns a score to every student response, and total scores are computed for individual models.

E. Evaluation Metrics

To compare the effectiveness and reliability of different models, multiple evaluation metrics were applied:

- Pearson and Spearman Correlation: Measures the strength of linear and rank-order agreement between model scores.
- Cohen's Kappa: Assesses inter-model agreement to determine consistency.
- Score Distribution Analysis: Uses Kernel Density Estimation (KDE) to visualize scoring patterns across models.
- Semantic Sensitivity Analysis: Evaluates how stable the models are when grading paraphrased responses. To assess the robustness of models to paraphrasing, a semantic sensitivity test was conducted. A set of paraphrased responses was created for three key questions while preserving their original meaning. Each model computed semantic similarity scores between the paraphrased responses and the corresponding correct answers. Higher scores indicate greater stability in recognizing reworded but semantically equivalent answers. This analysis evaluates the model's ability to generalize beyond exact phrasing, ensuring consistency in grading.
- Processing Time Analysis: Measures computational efficiency of each model.

IV. Results and Discussion

The evaluation of the four transformer-based models (MiniLM, BERT, RoBERTa, and SBERT) for grading student responses revealed significant variations in their performance. Table I presents the Pearson correlation, Spearman correlation, and Cohen's Kappa scores, which measure the similarity of total scores assigned by different models.

Pearson and Spearman correlations indicate the degree of linear and rank-based agreement between models. The highest correlation values were observed between MiniLM and SBERT (Pearson: 0.909, Spearman: 0.908) (Table I), suggesting that these two models assigned highly similar scores. RoBERTa and SBERT also exhibited strong correlation (Pearson: 0.799, Spearman: 0.795), indicating comparable scoring patterns. In contrast, the lowest correlation was observed between BERT and RoBERTa (Pearson: 0.720, Spearman: 0.689), suggesting that these models differed substantially in their scoring approach. The strong correlation between SBERT and MiniLM supports the inclination that SBERT effectively captures semantic similarity, likely due to its specialized sentence representation training.

TABLE I. Performance evaluation results of MiniLM, BERT, RoBERTa, and SBERT based on correlation and agreement analysis

Model Comparison	Pearson Correlation	Spearman Correlation	Cohen's Kappa
MiniLM vs BERT	0.742	0.752	0.026
MiniLM vs RoBERTa	0.765	0.759	0.094
MiniLM vs SBERT	**0.909**	**0.908**	0.323
BERT vs MiniLM	0.742	0.752	0.026
BERT vs RoBERTa	0.720	0.689	**-0.020**
BERT vs SBERT	0.745	0.745	0.147
RoBERTa vs MiniLM	0.765	0.759	0.094
RoBERTa vs BERT	0.720	0.689	**-0.020**
RoBERTa vs SBERT	0.799	0.795	0.017
SBERT vs MiniLM	**0.909**	**0.908**	0.323
SBERT vs BERT	0.745	0.745	0.147
SBERT vs RoBERTa	0.799	0.795	0.017

Cohen's Kappa assesses inter-model agreement beyond random chance [39]. The highest agreement was found between MiniLM and SBERT (0.323), followed by BERT and SBERT (0.147). In contrast, BERT and RoBERTa exhibited the lowest agreement (-0.020), implying near-random agreement levels. These findings suggest that some models exhibit more consistency in their grading tendencies, while others may rely on different linguistic cues when assigning similarity scores. The lower agreement scores for RoBERTa highlight its distinct processing characteristics compared to the other models. BERT and RoBERTa demonstrate near-random agreement (-0.020), as reflected in Cohen's Kappa scores which indicates inconsistencies. The near-random agreement between BERT and RoBERTa (-0.020 Cohen's Kappa) likely stems from differences in their pretraining objectives, contextual sensitivity, and score variance. RoBERTa's dynamic masking strategy and extensive training may enhance its semantic understanding, while BERT's reliance on next-sentence prediction (NSP) could limit its ability to capture nuanced relationships, leading to divergent grading patterns. Additionally, RoBERTa's higher score variance suggests a broader grading distribution, further reducing alignment with BERT.

To further analyse the differences among models, kernel density estimation (KDE) plots were used to visualize score distributions. The KDE plot as shown in Fig. 1 indicates that MiniLM and SBERT exhibited similar score distributions, reinforcing their high correlation. RoBERTa demonstrated a more concentrated distribution with distinct peaks, suggesting that it tends to assign scores within specific ranges. BERT, on the other hand, displayed a left skewed score distribution, indicating greater variation in its scoring patterns. The KDE visualization (Fig. 1) also shows that RoBERTa exhibits a more scattered distribution. This suggests that RoBERTa is more variable in its grading, potentially contributing to its lower agreement scores with other models.

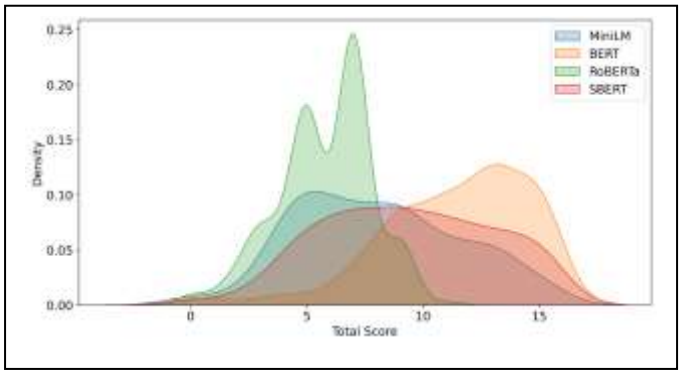


Fig. 1. Score distribution and variability of MiniLM, BERT, RoBERTa, and SBERT

A standard deviation analysis of scores per response revealed an average variability of 2.719 across models. This suggests that while all models capture semantic similarity, there are notable differences in their sensitivity to paraphrasing and lexical variations. The lower variability of SBERT supports its effectiveness in recognizing semantically equivalent answers despite surface-level differences.

To assess the robustness of models against paraphrased responses, a semantic sensitivity test was conducted as described in the section on evaluation metrics. SBERT consistently produced the highest similarity scores when comparing paraphrased responses to the correct answers, followed by BERT and MiniLM (Table II). RoBERTa demonstrated the lowest sensitivity, indicating that it may be more reliant on exact phrase matching rather than deeper semantic understanding. Specifically, as it can be observed in Table II, SBERT showed the highest sensitivity scores across all questions (Q1: 0.726, Q2: 0.881, Q3: 0.798), followed by BERT (Q1: 0.747, Q2: 0.834, Q3: 0.842). MiniLM performed moderately well, while RoBERTa lagged behind. These results reinforce SBERT's fine-tuned ability to discern contextual meaning beyond lexical overlap.

TABLE II. SEMANTIC SENSITIVITY SCORES FOR THE FOUR TRANSFORMER MODELS

Model	Q1 Score	Q2 Score	Q3 Score
MiniLM	0.673	0.821	0.776
BERT	0.747	0.834	0.842
RoBERTa	0.568	0.772	0.719
SBERT	0.726	0.881	0.798

The processing efficiency of each model was rigorously assessed, highlighting significant differences in execution time, as shown in Fig. 2. MiniLM exhibited the highest efficiency, completing the grading process in just 80.06 seconds, making it the most practical choice for real-time applications. SBERT, while demonstrating strong semantic sensitivity, required 155.56 seconds, making MiniLM approximately 1.94 times

faster. BERT's processing time of 405.01 seconds means MiniLM is about 5.06 times faster, placing BERT at a disadvantage for large-scale grading tasks. RoBERTa, with the longest execution time of 1359.64 seconds, is nearly 16.98 times slower than MiniLM, presenting a major computational bottleneck that limits its feasibility in time-sensitive environments. These variations in processing time emphasize the critical need for balancing efficiency and accuracy in automated grading systems, particularly in large-scale educational settings where timely feedback is essential.

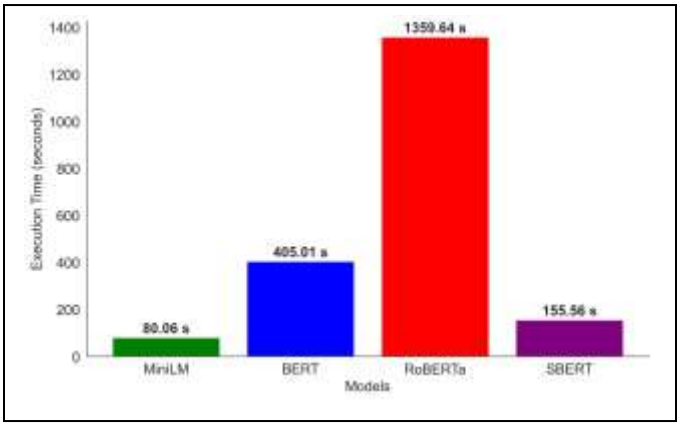


Fig. 2. Comparison of processing efficiency of the transformer models, showing execution time (in seconds) for MiniLM, BERT, RoBERTa, and SBERT models.

The results indicate that SBERT provides the best balance between semantic understanding, correlation with other models, and processing efficiency. MiniLM also emerges as a competitive alternative, offering a lightweight yet effective grading approach. However, the lower agreement scores between RoBERTa and other models suggest that it may require further fine-tuning for optimal performance in semantic similarity tasks.

V. Conclusion

This study demonstrated the effectiveness of transformer-based models in automating the grading of open-ended questions, addressing key challenges of subjectivity, inconsistency, and inefficiency in manual assessment. By evaluating MiniLM, BERT, RoBERTa, and SBERT on multiple grading metrics, SBERT was identified as the most robust model for capturing semantic similarity with high sensitivity and strong correlation to human grading. MiniLM, while slightly less sensitive, provides a practical trade-off between efficiency and accuracy, making it a viable option for large-scale applications.

The comparative analysis highlights the significant variations in model performance, particularly in agreement and computational efficiency. RoBERTa, despite its complexity, exhibits the highest processing time and variance in score distribution, making it less suitable for real-time grading. Meanwhile, the poor agreement between BERT and RoBERTa underscores the need for careful model selection to ensure reliable and consistent automated scoring.

These findings underscore the transformative potential of AI-driven grading systems in educational settings. By leveraging transformer-based models like SBERT and MiniLM, institutions can reduce the manual grading burden, enhance grading consistency, and improve the fairness of student assessments. Future work will explore how human-expert evaluations could be incorporated to benchmark automated scoring against human grading, further validating model effectiveness. Fine-tuning of models and testing on larger datasets are also significant areas to explore in future work.

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